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SUPERSEDING
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MILITARY SPECIFICATION

WELDING MACHINE, ULTRASONIC, SPOT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers ultrasonic welding machines of the spot type, which have the primary function of joining metal components (wire or sheet) with equipment and accessories specified herein.

1.2 Classification. The machine shall be of the following sizes. The size to be furnished shall be as specified (see 3.5 and 6.2).

Size	Electrical Power (watts to transducer)
2a	350
3a	700
4a	900
5	1000
5a	1500
6	2000
6a	2500
7	3000
7a	4500
7b	5000
9	6000

Beneficial cormnents (recomnendatlons, add It Ions, deletlons) and any pertlnent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS) 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards,

(Application for copies should be addressed to the Superintendent of Documents', U. S. Government Printing Office, Washington, DC 20402-0001.)

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which" are DOD adopted are those listed in-the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B93.114M - Pneumatic Fluid Power - Systems Standard for Industrial Equipment and Machine Tool Applications.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, Ny 10(118-3363.

AMERICAN WELDING SOCIETY (AWS)

AWS C1.1 - Welding, Resistance, Recommended Practices for.

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125-3136.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

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3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall be designed to convert electrical energy to mechanical vibratory energy. The vibratory energy shall be transmitted through the welding head to the work components and a predetermined pressure applied for a specific time interval to accomplish the ultrasonic spot weld. The machine shall be designed to operate in ambient temperatures ranging from 41° F to 1040 F and relative humidities ranging from 20 percent to 95 percent non-condensing. All components, parts, and features necessary to meet the performance requirements specified herein shall be included in the machine. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated In INCH-POUND (see 6.5.1), metric (see 6.5.2), or dual (INCH-POUND and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate pressure shall be permanently and legibly engraved or etched on a nonglare background and shall be graduated in the smallest increments that can be easily read from the operator's normal position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possibl e provided such materials will not jeopardize the intended use, performance, or" design life of the machine. Recl~imed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy In normal operation shall be designed and constructed for energy efficiency in accordance with the developme~ts available within the industry.

3.2.4 Controls. All mechanical, electrical, and pneumatic operating controls shall be located convenient to the operator's work station.

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assembles furnished on he machine, whether or not specifically required herein, shall conform with all requirements of 29 CFR 1910. If a conflict arises between 29 CFR 1910 and AWS C1.1, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified in detail (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the cormnercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. Additional guarding dependent on workpiece size and configuration shall be provided when specified and fully described by the procuring activity (see 6.2).

3.2.6 Mercury restriction. The machine shall not fcontain mercury or mercury compounds nor be exposed to free mercury during manufacture.

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3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be u,sed on or in he machine.

3.2.8 Environmental protection. During the manufacture, operation, service, transportation, or storage of the machine, and its components materials hazardous to the ecological system as prohibited by Federdl, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine. All replacement parts shall be interchangeable with their respective original parts, and fit/function with mating components.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion, or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where speclfled in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Paintin Unless otherwise specified (see 6.2), the machine shall be painte+ In accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachmen~accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

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3.3.7 Electromagnetic interference control. When specified (see 6.2), equipment furnished under hls specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than an ultrasonic frequency converter, ultrasonic welding head, electrical system, and a pneumatic system.

3.4.1 Ultrasonic frequency converter. The ultrasonic frequency converter shall provide he necessary welding voltage, current, and frequency characteristics when connected to the standard commercial power source specified herein. All components shall be secured to preclude damage from shock stresses, vibration, or other conditions encountered during service, operation, shipping, and storage.

3.4.1.1 Converter ultrasonic power output. The frequency converter shall provide ultrasonic welding power as required for the size machine specified. Control of power output shall be variable. Power output of the converter shall vary not more than 1.0 percent at any setting. A power indicating meter for power calibration shall be provided.

3.4.1.2 Converter frequency. The frequency converter shall power the transducer-coupling assembly at a fixed frequency as specified by the manufacturer. An indicating meter shall be incorporated to facilitate adjustment of nominal operating frequency.

3.4.1.3 Converter weld pulse duration. Each power unit shall be equipped with an adjustable weld pulse timer. The range of weld pulse duration, from minimum to maximum, shall be as specified by the manufacturer. Pulse duration repeatability shall be within +0.01 second. All timers shall be standardized at 0.1 second +0.01 second. 'A set up mode of operation to continuously activate the uni~ shall be provided.

3.4.2 Ultrasonic welding head. The ultrasonic welding head shall include a welding tlp and a transducer-coupling assembly consisting of a transducer, a coupler, and a mounting head. The head shall be designed for the application of an adjustable and repeatable ultrasonic welding force. The time and force applied by the head shall be repeatable to within ±5 percent of the preselected setting.

3.4.2.1 Weldin ti The welding tip shall be mounted *on* the coupler and shall transml t e v1 'fatory welding energy and force to the weldment.

3.4.2.2 Transducer-coupling assembly. The transducer-coupling assembly shall provide mechanical vibratory energy by means of a transducer and coupler to the welding tip. The frequency shall be in accordance with the manufacturer's standard commercial practice for the size specified (see 1.2). Frequency shall vary by not more than ~1.0 percent.

3.4.2.2.1 Transducer. The transducer shall convert high frequency electrical energy to mechanical vibratory energy of the same frequency. The transducer shall be rigidly attached to the coupler and shall minimize acoustic wave distortion or attenuation during transmission into the coupler.

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3.4.2.2.2 Cou ler. The coupler assembly shall transmit ultrasonic energy to the weldlng lp.+ The construction of the coupler shall prevent the generation of undesirable acoustical characteristics. The coupler shall be mounted to isolate the ultrasonic system from the machine frame under all operating conditions.

3.4.2.2.3 Mountin head. The mounting head shall hold the welding tip. The mounting~ea acilitate changing the welding tip with a minimum loss of time and minimum effort. The mounting head shall be rigid and shall ‘ Dermit no detrimental mechanical deflection of the weldina tiD when the normal i-ange of static clamping force is applied.

3.4.3 Work-holding unit. When specified (see 6.2), the work-holding unit shall be constructed of materials that are compatible with ultrasonic welding processes and shall be generally non-compliant at the welding frequency and contraresonant.

3.4.4 Electrical requirements. The machine shall be wired to conform to NFPA”79. Unless otherwise spec~fied [see 6.2), sizes 2a, 3a, 4a, 5. and 5a shall operate on single phase, 60-Hertz (Hz), ’11O or 220-volts; sizes 6, 6a, and 7 shall operate on single phase, 60 Hz, 220 volts; and sizes 7a, 7b, and 9 shall operate on 3-phase, 60 Hz, 480 volts.

3.4.4.1 Controls. The controls for the welding machine shall provide for control of all components and functions of the machine. All control and indicating devices on the machine shall be mounted and located within sight and reach of the operator when in his normal operating position. Each control and indicator shall be clearly and legibly marked for function and identification. The controls shall operate on not more than 120 volts, single phase, 60 Hz input power. The power supplied shall vary not more than ±10 percent.

3.4.4.1.1 $lanual control. Unless otherwise specified (see 6.2), machine sizes 2a, 3a, 4a, 5, 5a, 6, 6a, and 7 shall be supplied with manual controls with standard, basic electrical components.

3.4.4.1.2 Programmable control. Unless otherwise specified (see 6.2), machines sizes 7a, 7b, and shall be supplied with a prograniwnable control. The prograniunable control shall control all machine functions necessary to ultrasonically weld the workplaces automatically, by storing program information in a non-volatile memory. Manual setup shall be accomplished by a keylock switch.

3.4.5 Pneumatic system. The pneumatic system shall conform to the requirements of ANSI B93.114M. Unless otherwise specified (see 6.2), the machine shall be designed to operate on a pressure not greater than 100 pounds per square inch gage (psig) of clean dry air and a volume of not less than 3 cubic feet per minute (cfm).

3.4.6 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shaTl be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of O to 9,999 hours in increments of

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not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. ***Unless*** otherwise specified (see 6.2), the size and capacity of he machlrte shall meet the requirements of table I for the size specified. Machine size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. The overall envelop of the machine shall be limited, when specified (see 6.2). The physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its operating location.

3.6 Performance. The machine shall be capable of performing ultrasonic spot weld lnq operations of metal components (wire or sheet) in the workpiece thicknesses-specified in table I for"the size-machine required.

3.6.1 Additional performance requirements. Performance features required in addition to, or In lleu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's corrrnerclal_ machine shall be furnished.

3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or machine.

Required repair parts shall be furnished as spec"ified ($ ~ ' 6 . _ *

3.10 Marking on instruments, control panels, charts, and plates. All !Worals on instruments. control t)anels. charts. and ~lates shall be In the Emglish language. Cha~acters shail be permanen~ly marked in boldface on a contra~ting background. All plates shall be corrosion resistant.

3.10.1 Lubrication plate. A corrosion resistant lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below:

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each
Lubricant (if applicable)

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TABLE I. Ultrasonic spot welding - rated capacity.

MATERIAL	ALLOY	Material thickness (inches), range 1/					
		Size 2a	Size 3a	Size 4a	Size 5	Size 5a	Size 6
		350 watts 2/	700 watts 2/	900 watts 2/	1,000 watts 2/	1,500 watts 2/	2,500 watts 2/
		using .250 diameter welding tip	using .250 diameter welding tip	using .250 diameter welding tip	using .250 diameter welding tip	using .250 diameter welding tip	using .250 diameter welding tip
Aluminum	1100-H14 3003-H14	.001 - .008	.001 - .015	.001 - .018	.001 - .020	.001 - .030	.001 - .045
Copper	Soft	.001 - .008	.001 - .015	.001 - .018	.001 - .020	.001 - .030	.001 - .040
Copper	Half-hard	.001 - .010	.001 - .015	.001 - .020	.001 - .020	.001 - .025	.001 - .030
Brass	70Cu:30Zn	.001 - .020	.001 - .030	.001 - .032	.001 - .032	.001 - .035	.001 - .038
Be Copper	CU-2.25Be	.001 - .004	.001 - .008	.001 - .010	.001 - .012	.001 - .016	.001 - .020
Cupro-Nickle	90Cu:10nl	.001 - .008	.001 - .015	.001 - .018	.001 - .020	.001 - .025	.001 - .032
		using 3/ welding tip	using 3/ welding tip	using 3/ welding tip	using 3/ welding tip	using 3/ welding tip	using 3/ welding tip
Copper Wire	Stranded	.001 Dia.- CMA.12 4/	.001 Dia.- CMA.25 4/	.001 Dia.- CMA.35 4/	.001 Dia.- CMA.38 4/	.001 Dia.- CMA.50 4/	.001 Dia.- CMA.62 4/
Copper Wire	Magnet Wire	.005 Dia.- AWG 22 5/	.005 Dia.- AWG 18 5/	.005 Dia.- AWG 16 5/	.005 Dia.- AWG 16 5/	.005 Dia.- AWG 14 5/	.005 Dia.- AWG 14 5/

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IABLE 1. Ultrasonic #pot welding - ratd~.~~~ - Continued.

WKBIAL	ALLOT	Uaterial thickness (inched, range ~/ Size 6a ~500 matta 2J using .250 diameter mlding tip	Size 7 -ilooo mttg V wing ,250 diameter welding tip	Size 7a ~L~-itt6 _21 using .250 diameter welding tip	Size 'lb -~o watts 21 using .250 diameter welding tip	Size 9 -6,000 wattg ~ uuing .250 diamater wdding tip
Aluminua	11 OO-H14 3003-H14	.001 - .062	.001 - .079	.001 - .100	.001 - .105	.001 - .110
Copper	soft	.001 - .050	.001 - .064	.001 - .078	.001 - .082	.001 - .087
Copper	51 f -h*I'd	,001- .035	.001 - ,042	,001- .050	.001 - .053	.001 - ,055
6MM	70CU:30Z0	,001- .040	,001- .040	.001 - .040	.001 - ,042	.001 - .044
Be copper	CU-2 . 25Be	.001 - .025	.001 - .028	.001 - .032	.001 - .034	.001 - .035
Cupro-JickIe	90Cu: 10nl	.001 - .040	.001 -.045	.001 - .050	.001 - .053	.001 - .055
		wing :/ mld~ tip	wing ~/ welding tip	'-wing j/ -welding tip	using ~L melding tip	using # welding tip
Copper Wire	Stranded	.001 Dia. - CUA.75 y	.001 Dia. - CUA.88 J/	.001 Dia. - ARI 4 ~1	.001 Dia. - AUI 4 j/	.001 Dia. - AE O ~J
@pLr Wire	Maanet Wire	.005 Dia. - AM 12 51	.005 Dia. - AM 12 5/	.005 Dia. - AW 10 5/	.005 Dia. - Am6 51	.005 Dia. - AIE 4 51

IO'm:
~/ Y'hicknea of upper wkpiece.
~/ Electrical power (natt# to transducer).
3/ Ueldin6 tipg #hall be provided aa required to ensure an acceptable ultrasonic itpot-oeld for all wire #ize# within the range listed for each machine #iz@.
~/ CM is an abbreviation for circular mill area.
~/ Am ig an abbreviation for Amrican Wire gage,

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3.10.2 Nam& late." A corrosion resistant nameplate shall be securely attached to eac"machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.11 Workmanship . Workmanship of the machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it conrnit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

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4.3 Inspection conditions. Unless otherwise specified (see 6.2), all insr)ectlons shall be Derformed in accordance with the test conditions spe;ified in 4.8.1. ‘

4.4 First article inspection. When a first article inspection is required, it shall be amlled to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.3. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. An acceptance test shall be performed on each machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each machine shall be examined to determine compliance with al? requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be ~erformed in an indoor facility with ambient conditions of 410 to 1040 Fahlrenheit (F) and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. Each machine sha’l1 be operated at no load for not less than one hour. Proper operation of all controls, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Performance tests. The machine shall perform ultrasonic spotwelding at the capacities llsted ip table I. Unless otherwise specified (see 6.2), the machine’s performance shall be tested using aluminum alloy 11OO-H14 or 3003-H14 test samples. When specified (see 6.2.), the machine’s performance shall be tested using materials other than aluminum test samples (see table I). Test materials shall be clean. Material size for test samples shall be 2.5 inches wide, 7.5 inches long, and the maximum thickness shown in table 1 for the machine size ordered (see 1.2). Test sample material size for the minimum capacity shall be 1 inch wide, 5 inches long, and the thickness shown in table I for the machine size ordered (see 1.2). The ends of the test sample material shall be overlapped 3 inches and the welds shall be centered. Ten ultrasonic spot-welds, .25 inch apart shall be performed on each weldment. Seven weldments shall be fabricated. Visual examination of the test welds under 30 power magnification shall reveal no cracks and the combined area of voids shall be not greater than 10 percent of the interface contact area. The weld samples shall be tested by pulling the material apart in tension. There shall be no weld separations and the material shall tear” at the nuggets.

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4.8.3.1 Additional% performance tests. Performance tests required in addition to or in' lieu cff the requlremenfs of 4.8.3 shall be as specified and fully described (see 6.2).

4.8.4 Ultrasonic characteristics tests. Ultrasonic probe measurements on the work holder shall show vibratory amplitude to be down to not more than ten-to-one from welding-tip amplitude. Resonant frequency shall be established with the transducer-coupling assembly unloaded and with 11OO-H14 or 3003-H14 material specimens under the welding tip.

4.8.5 Electromagnetic interference control test. When specified (see 6.2), equipment requlrlng electromagnetic interference control shall be tested for compliance with 3.3.7 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance wlth MIL-W-45562.

6. NOTES

6.1 Intended use. The welding machine covered by this specification is intended for use in producing ultrasonic spotwelds under production conditions in accordance with this specification in the materials and thicknesses indicated herein.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Specify system of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g" Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different (see 3.3.5.).

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i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.7).

j. Work holding unit, if required (see 3.4.3).

k. Electrical requirements, if different (see 3.4.4).

1. Specify controller, if different [see 3.4.4.1.1).

m. Specify controller, if different (see 3.4.4.1.2].

n. Pneumatic requirements, if different (see 3.4.5).

o. Hour meter, if required (see 3.4.6).

p" Size and capacity, if different (see 3.5).

q' Specify physical size limitation, if required (see 3.5.1).

r. Additional performance features, if required (see 3.6.1).

s. Specify and fully describe optional equipment, if required (see 3.8).

t. Specify applicable repair parts, if required (see 3.9).

u. Specify information to be provided on lubrication plate if different (see 3.10.1).

v. Nameplate, if different (see 3.10.2).

w. Inspection conditions, if different (see 4.3).

x. First article inspection, if different (see 4.4).

y. Quality conformance inspection, if different (4.5).

z. Acceptance test, if different (see 4.6).

aa. Material to be used for performance test, if different (see 4.8.3).

bb. Other test materials, if required (see 4.8.3).

cc. Additional performance tests, if required (see 4.8.3.1).

dd. Electromagnetic inference control test, if required (see 4.8.5).

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6.3 Consideration of data requirements. The following data requirements should ~e cons~dered when his specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the remarks section of the DD Form 1423 hat alr technical data furnished should be written in the English language.

6.4 First article. When first article inspection is required, the contracting off"leer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. "

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound corrrnonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (S1)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

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6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to met~ic units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification. Classification changes from the previous issue of his specification as listed in table II.

TABLE II. Cross-reference sizes of ultrasonic spot welding machines.

MIL-W-80227	MIL-W-80227A
Size 1	Deleted
Size 2	Deleted
	Size 2a (new size)
Size 3	Deleted
	Size 3a (new size)
Size 4	Deleted
	Size 4a (new size)
Size 5	Size 5
	Size 5a (new size)
Size 6	Size 6
	Size 6a (new size)
Size 7	Size 7
	Size 7a (new size)
	Size 7b (new size)
Size 8	Deleted
Size 9	Size 9

6.8 Chanaes from mevious issues. Marainal notations are not used in this revision to ldentlfy changes with respec~ to the previous ssue due to the extensiveness of the changes.

6.9 Safetv and health requirements. The 29 CFR 1910 limil s only the total hazard level (noise, radlatlon, electromagnetic emissions, noxious-vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements ***necessary*** to integrate this new machine into its future environment.

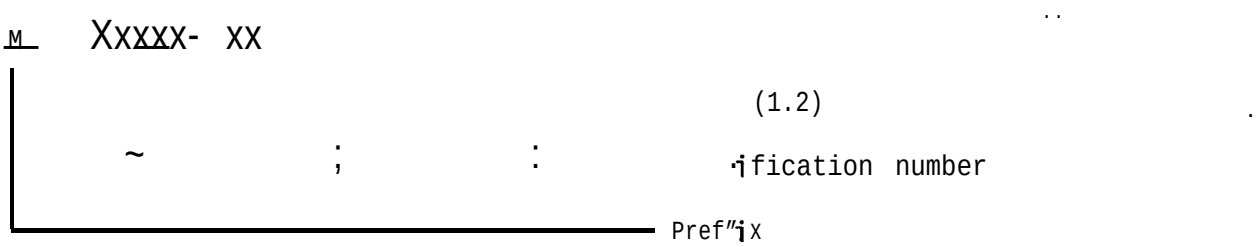
p;;;;em;fd;[a]; ~[i];Z~;~~ ~~ZZt? ‘he ‘rocuring activity ‘hOuld be

6.11 Warranty. Warranty requirements should be as specified by the procuring actlvlty in the contract.

6.12 Inspection location. The ***contractor*** should identify in his response to the solicitation the location where inspection and tests are to be performed.

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6.13 Part or Identifying Number (PIN). The PIN to be used for ultrasonic spot-welding machines acquired to this specification are created as follows:



6.14 Subject term (key word) listing.

High frequency welding
Ultrasonic joining
Vibratory welding

Custodian:
Air Force - 99

Review activity:
Air Force - 84

User activity:
Air Force - 71

Preparing activity:
DLA - 1P

Project 3438-0066

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-W-80227A	2. DOCUMENT DATE (YYMMDD) 21 February 1990

3. DOCUMENT TITLE
WELDING MACHINE, ULTRASONIC, SPOT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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